Robust Deepfake Detection Under Real-World Perturbations

Rajesh Kumar
Dept. of CSE, Sahyadri College of Engineering and Management Mangaluru, India rajesh.kumar@example.edu

***Abstract*—This paper studies robust detection of manipulated face videos with a lightweight network. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.**

***Index Terms*—deepfake detection, forensics, robustness**

I. Introduction

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. In our previous work [1], we have recently proposed a detector, and [2] studied this. Code: https://github.com/rajesh-lab/deepfake-det, see also [3]–[5]. Results are in Table I, Fig. 1, Fig. 2, Eq. (1) and Eq. (2); see Section III.

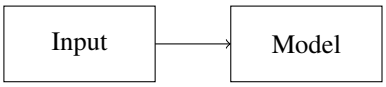


Fig. 1. Overview of the proposed pipeline.

TABLE I
Comparison with prior methods

Method	Acc	AUC
hline Baseline	81.2	0.85 Ours
93.7	0.96	
hline		

II. Method

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. The loss is

$$L = \sum_{i} (y_i - \hat{y}_i)^2 \tag{1}$$

and the score is

$$s = \sigma(w^T x + b). \tag{2}$$

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

III. Experiments

Deepfake detection has become an important problem in multimedia forensics because modern generators produce

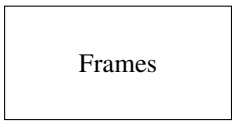


Fig. 2. Example frames.

faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

IV. Conclusion

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

Acknowledgment

This work was supported by the DST-SERB grant no. 1234.

References

[1] A. One, “Paper one,” *IEEE Trans. Inf. Forensics Security*, vol. 1, 2020.
[2] B. Two, “Paper two,” in *Proc. CVPR*, 2021.
[3] C. Three, “Paper three,” in *Proc. ICASSP*, 2022.
[4] D. Four, “Paper four,” *IEEE Access*, 2023.
[5] E. Five, “Paper five,” in *Proc. ICIP*, 2024.